NOT MEASUREMENT SENSITIVE

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SUPERSEDING
MIL-S-45152G
16 December 1983

MILITARY SPECIFICATION

SEMITRAILERS, LOWBED: COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers commercial lowbed semitrailers. Each vehicle procured under this specification is a commercial item which shall be warranted by the manufacturer as specified in acquisition documents.

1.2 Classification. The semitrailer shall be one of the following types, classes, sizes, and styles, as specified (see 6.2).

Type I	- Level deck
Type II	- Drop deck
Type III	- (see 6.4)
Type IV	- (see 6.4)
Type V	- Level deck extended

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document or by letter.

AMSC N/A

FSC-2330

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	Class A	- Fixed gooseneck
	Class B	- Folding gooseneck, for use with a truck tractor mounted winch
	Class C	- Detachable gooseneck
	Class D	- Folding gooseneck, hydraulic, self-contained
*	Class E	- Detachable gooseneck, hydraulic, self-contained
	Size 15	- 15 ton
	Size 20	- 20 ton
	Size 25	- 25 ton
	Size 35	- 35 ton
	Size 50	- 50 ton
	Size 60	- 60 ton
	Size 75	- 75 ton
	Style 4W	- 4 wheel
	Style 6W	- 6 wheel
	Style 8W	- 8 wheel

1.2.1 Combination. The combinations of type, class and style ordered under this specification shall be limited to those shown in table I.

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TABLE I. Permissible combinations.

Type	Class	Size and style												
I	A	15-4W	20-4W	25-4W	35-4W	35-6W	50-6W	50-8W	60-6W	60-8W	75-6W	75-8W		
	B,D				35-4W	35-6W			60-6W	60-8W				
II	B,D								60-6W	60-8W	75-6W	75-8W		
	C			25-4W	35-4W	35-6W	50-6W	50-8W	60-6W	60-8W	75-6W	75-8W		
V	A		20-4W	25-4W										

NOTE: Types III and IV no longer applicable (see 6.4).

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DODDIS), and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

W-B-131	- Battery Storage: Vehicular, Ignition, Lighting and Starting.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-W-3912	- Wood Parts, Fabricated, For Transport Vehicle Bodies and Similar Applications.

STANDARDS

FEDERAL

FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-1223	- Non-Tactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
MIL-STD-1261	- Arc Welding Procedures for Constructional Steels.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS 75021	- Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 24-Volt, Waterproof.

* (Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DEPARTMENT OF DEFENSE
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government printing office for hard copy, or microfiche copies are available from the Director, Navy Publications and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

* AMERICAN WELDING SOCIETY (AWS)
Structural Welding Code, Steel, D1.1

(Application for copies of AWS publications should be addressed to the American Welding Society, 2501 NW 7th Street, Miami, FL 31025.)

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of the European Tyre and Rim Technical Organisation publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SAE, INC.
SAE Standards and Recommended Practices

J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Truck, Truck-Tractors, and Trailers (DoD adopted).
J517	- Hydraulic Hose.
J534	- Lubrication Fittings.
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J588	- Turn Signal Lamps.

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J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
J700	- Fifth Wheel Kingpin - Commercial Trailers and Semitrailers (DoD adopted).
J702	- Brake and Electrical Connection Locations - Truck-Tractor and Truck-Trailer (DoD adopted).
J1292	- Automobile, Truck, Truck-tractor, Trailer, and Motor Coach Wiring.

(SAE Standards and Recommended Practices are available from SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book

(Application for copies of Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the document. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.7, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be the manufacturer's standard or optional items, which meet or exceed the requirements of this specification. All vehicle items shall be as represented and rated in the vehicle and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components and shall be on file in the engineering offices of the procuring activity prior to delivery of the items.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

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* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

* 3.1.1.2 Marking and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223. When specified (see 6.2), concealed markings shall be furnished.

* 3.1.1.3 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297.

3.1.1.4 Wood treatment. Wood, except apitong, shall be treated in accordance with MIL-STD-1223.

* 3.1.1.5 Additional data plates. Additional data plates depicting the maximum towing speeds, permissible loads and permissible load distributions (see 3.3 through 3.4), shall be provided. The plates shall indicate the maximum towing speeds with and without payload, the maximum concentrated payload for each four foot segment of the entire length of the semitrailer, including the gooseneck, and the longitudinal location, length and width(s) of the maximum concentrated loads permissible as specified in 3.3. Appropriate loading instructions and cautions, as applicable to the manufacturer's design, shall be furnished on the plates. A plate shall be provided on each side of the vehicle in a conspicuous location. The plates shall conform to composition A (class 1 or 2) or composition C of MIL-P-514.

* 3.1.1.6 Lifting and tying down attachments. When specified (see 6.2), the semitrailer shall be equipped with lifting and tying down attachments. Lifting and tying down attachments shall conform to type II or type III of MIL-STD-209. A transportation plate conforming to composition A (class l or 2) or composition C of MIL-P-514 shall be attached to the semitrailer. The transportation plate shall be inscribed with a diagram showing the lifting attachments and proposed lifting slings, the capacity of lifting attachments and proposed lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the semitrailer, showing the center of gravity, shall be provided on the transportation plate. Tiedown attachments shall be identified by stenciling or other suitable markings. Tiedown markings shall clearly indicate that the attachments are intended for the tiedown of the semitrailer on the carrier when shipped.

* 3.1.1.7 Lubrication chart. When specified (see 6.2), a lubrication chart shall be provided. The chart shall direct attention to all lubrication fittings and specify the range and grade of lubricants required for critical temperatures. The chart shall be permanently attached to the semitrailer in a readily visible location. The chart shall be inscribed on a plate conforming to composition A (class l or 2) or composition C of MIL-P-514.

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3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle shall comply with all Federal Motor Vehicle Safety Standards applicable to the type of semitrailer furnished and in effect on the date of manufacture.

* 3.2.2 Dissimilar metals. All dissimilar metals used throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.

* 3.2.3 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be available to the appropriate engineering office of the procuring activity.

* 3.2.4 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.5 Asbestos. Asbestos materials shall not be used in any form in any part of the vehicle.

* 3.3 Performance. The fully equipped semitrailer shall evidence no part failure, deformation, permanent set or interference between parts when towed under each of the conditions specified in table II. The semitrailer also shall show no part failure, deformation, permanent set or interference between parts when towed on 15 percent side slopes, at speeds up to 10 mph, when loaded as specified in 3.4 and when empty.

TABLE II. Performance.

Condition	Payload	Speed (minimum)
Unimproved roads; other reasonably hard, uneven terrain	None	10 mph
Unimproved roads; other reasonably hard, uneven terrain	Rated payload	10 mph
Highways	None	Sizes 15, 20, 25, 35: 55 mph Sizes 50, 60, 75: 40 mph
* Highways	Rated payload	Sizes 15, 20, 25, 35: 55 mph Sizes 50, 60, 75: 30 mph

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3.3.1 Turning ability. The semitrailer shall be capable of assuming a 90 degree angle to a coupled truck tractor without cramping and without damaging the semitrailer or the truck tractor.

* 3.3.2 Brake performance. The service brakes shall stop the truck tractor-semitrailer combination, with the semitrailer loaded with specified payload, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52. Requirements shall be met with a truck tractor loaded to its full rated gross vehicle weight (GVW) and gross combination weight (GCW).

3.3.3 Tracking ability. The semitrailer shall conform to the tracking requirements of Federal Motor Carrier Safety Regulation 393.70(a).

3.3.4 Folding gooseneck performance. The gooseneck on class B and class D folding gooseneck semitrailers shall be capable of folding to the ground and of being raised and locked as specified herein, with the rated payload positioned as specified herein (see 3.4) without evidence of any type of component failure, deformation, permanent set, misalinement or interference between parts.

* 3.4 Payload and distribution. The semitrailer shall be capable of transporting the rated payload (shown in table III for the applicable size) without permanent deformation in the frame or gooseneck or any other part, under the operating conditions specified in 3.3, with the load:

(a) distributed as specified in table III for types I and II or
(b) evenly distributed in the loadspace for type V. In addition, type V semitrailer shall be capable of transporting 65 percent of rated payload concentrated (distributed) as specified in table II for the applicable size of semitrailer.
(c) Further, all types of semitrailers shall be suitable for the concentrated, pin point, and localized stresses caused by mobile loads being driven onto the vehicle, by tracks, track pads and wheels, and any other non-evenly distributed loads indicated by the full range of intended uses specified in 6.1.

TABLE III. Payload distribution.

Size	Rated payload, pounds	Distribution	
		Length of evenly distributed payload in center of load space, feet	Width of each crawler track, centered on load space, inches
15	30,000	8	16
20	40,000	10	22
25	50,000	10	22
35	70,000	11	24
50	100,000	11 1/2	30
60	120,000	12 1/2	32
75	150,000	14	34

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* 3.5 Dimensions and clearances. The semitrailer, without payload, uncoupled from the towing vehicle, resting on its landing legs on level ground, and with the platform level, shall conform to the dimensions specified in table IV and the following:

(a) Length of usable loadspace, feet (minimum) (NOTE: SEE 3.5.1)
For type I and II 16
For type V . . . . . 25 or 40, as specified (see 6.2)

(b) Upper fifth wheel height, from the ground to the underside of the upper fifth wheel plate, inches, plus or minus 1 inch
For size 15, 20, and 25 48
For size 35 52
* For size 50 and 60 56
* For size 75 58

(c) Distance from front of semitrailer to vertical centerline of kingpin, inches, plus or minus 2 inches
For type I and II. 18
For type V. 22

(d) Turning clearance, from the centerline of kingpin to any portion of semitrailer 6 inches or more below upper fifth wheel plate, inches (minimum) 82

(e) Forward swing clearance, from center of kingpin to most distant point on semitrailer nose, inches (maximum)
For type I and II 60
For type V 56

(f) Ground clearance, to bottom of frame rails, inches (minimum)
For type I and V 15
For type II 9

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TABLE IV. Dimensions and clearances.

Size and style	Dimensions										
	15-4W	20-4W	25-4W	35-4W	35-6W	50-6W	50-8W	60-6W	60-8W	75-6W	75-8W
Overall length, (feet, maximum)											
For type I and II	35	35	35	35	40	40	35	40	35	40	35
For type V		N/A	N/A								
Overall width, (inches, ± 1 inch)	95	95	95	95	95	95	95	95	120	95	140
Platform height, (inches, maximum)											
For type I	38	38	38	40	40	40	40	40	40	42	42
For type II			26	28	28	28	28	28	28	30	30
For type V		44	44								

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3.5.1 Load space. The usable load space of the platform shall be defined as follows:

(a) For type I level deck - Length from the rear of the gooseneck to the rear of the platform ahead of the tires
(b) For type II drop deck - Length from the rear of the gooseneck to the frame splices at the kickup over the wheels
(c) For type V level deck extended - Length from the rear of the gooseneck to the rear of the semitrailer.

3.6 Construction and stresses. Main frame members shall be adequately braced with a sufficient number of crossmembers to present a rigid construction with sufficient strength for the loads imposed. Except for type V level deck extended semitrailer, the outside frame members shall run the full length of the load space. The rear crossmembers on all types shall extend the full width of the semitrailer. When the semitrailer is loaded with rated payload, and with the loads distributed as specified in 3.3, the maximum fiber stress in each frame member(s) shall be not greater than 50 percent of the frame material yield strength or, when specified (see 6.2), 33 percent of the frame material yield strength. Under the above cited conditions of loading, the main frame deflection shall be not greater than 2 inches. All parts of the framing shall be completely electric-arc welded. The strength of the finished welds shall be such that they will not fail or crack when the semitrailer is subjected to the loads and performance requirements specified herein. On class B and class D folding gooseneck semitrailers, the underside, front and top of the gooseneck, shall be furnished with high strength, low alloy materials providing abrasion and corrosion resisting qualities. For class B, C, D and E folding and detachable gooseneck semitrailers, the maximum fiber stress, in any of the gooseneck area components, shall be not greater than 25 percent of the component material yield strength, with the rated semitrailer payload applied during loading, traveling, and unloading operations.

* 3.6.1 Frame splices. Where frame members are spliced, joints shall be reinforced to fulfill the dynamic loading requirements of 3.3 through 3.4 and the static loading requirements of 3.6. Splices shall be designed to avoid stress concentration. Splice welds shall be continuous. Transition areas in the gooseneck and kickup shall be smooth and rounded to minimize stress concentrations.

3.6.2 Stress analysis. A stress analysis for the semitrailer, loaded as specified in 3.4, shall be performed. The stress analysis shall include shear and moment diagrams and deflection calculations. The strength of the floor, crossmembers, and outside frame members shall not be included in calculations to compute the main frame maximum fiber stress; but the weight of the floor, crossmembers, and outside frame members imposed on the main frame shall be included in the total load imposed on the main frame. Stress calculations shall include complete analysis of the gooseneck, kickup area, floor, crossmembers, and longitudinal frame members.

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3.7 Platform.

* 3.7.1 Flooring. The platform shall be floored with apitong, white or red oak, or white ash of 2-1/4 inch uniform thickness as specified in MIL-W-3912. The tolerance on wood thickness shall be plus or minus 1/8 inch. The average weight of apitong used shall be not less than 50 pounds per cubic foot (pcf), with none of the individual pieces weighing less than 47 pcf, with no more than 15 percent moisture content at time of weighing. Flooring shall be laid lengthwise, as close together as practicable, attached with corrosion-resistant countersunk bolts and bridge style floor clips to eliminate drilling frame members. The top of countersunk bolt heads shall be not less than 1/8 inch and not more than 1/4 inch below the top of the wood surface. Exposed ends and edges of the flooring shall be protected with metal reinforcements. Wood surface shall be not less than 1/4 inch higher than adjoining metal members to prevent metal-to-metal contact of the load and the semitrailer. Platform crossmembers and bracing shall be adequate to sustain the rated payload with the load distributed as specified in 3.4. When platform width extensions are specified (see 3.7.9) the two center boards shall be secured to permit their easy removal by one man without special tools. The removable boards shall be capable of being used in the extension brackets specified.

* 3.7.2 Type I level deck. The platform shall be of the level deck type, extending from the rear of the gooseneck and continuing straight, with no wheel housings or kickup over the wheels. Wheelwells in the platform will be permitted; however, with the semitrailer on a level surface and with payload, and while the semitrailer is being loaded with a mobile payload from the rear, the tires shall not extend above the top of the platform.

3.7.3 Type II drop deck. The platform shall be of the drop deck type with a drop in the gooseneck area and with a kickup over the rear wheels.

* 3.7.4 Type V level deck extended. The platform shall be completely covered and level, extending from the rear of the gooseneck to the rear of the semitrailer with no protrusions or cutouts. If wheelwells are necessary, they shall be covered with steel plates. Under all conditions, there shall be no interference between the tires and the platform or wheelwell covers when the semitrailer is loaded with the specified payload and subject to the maximum suspension system articulation.

3.7.5 Platform extension (beavertail). When specified (see 6.2), a platform extension (beavertail) shall be provided. The length of the beavertail shall be a nominal 24 inches with a slope not less than 12 degrees and not more than 15 degrees. When a beavertail is furnished, the maximum specified overall length of the semitrailer may be increased by not more than 2 feet.

3.7.6 Stake pockets. When specified (see 6.2), type I level deck and type II drop deck semitrailer shall have eight stake pockets, four on each side. Unless otherwise specified (see 6.2), type V level deck extended semitrailer shall have stake pockets as follows:

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twelve stake pockets, six on each side for the 25-foot platform; sixteen stake pockets, eight on each side for the 40-foot platform. Stake pockets shall be equally spaced from front to rear of the loading space, except when interference with critical frame members occurs. The stake pockets shall be 4 inches square and shall be located inside the inner edge of the outer rails of the loading space. Stake pockets shall be furnished with drains to prevent entrapment of water.

3.7.7 Lashing D rings Lashing D rings shall be the heavy duty type and shall not protrude beyond the outer edge of the platform when not in use. Types I level deck and type II drop deck semitrailers shall have ten lashing rings, located as follows: three on each side of the platform; one on each side of the gooseneck, inside the outer edges; and two across the rear. Type V level deck extended 25-foot semitrailer shall have twelve lashing rings, located as follows: four on each side of the platform; one on each side of the gooseneck, inside the outer edge; and two across the rear. Type V level deck extended 40-foot semitrailer shall have fourteen lashing rings, located as follows: five on each side of the platform; one on each side of the gooseneck, inside the outer edge; and two across the rear. Each lashing ring and its mountings shall be capable of withstanding pulls of 10,000 pounds in any direction without permanent deformation.

3.7.8 Rails and pockets for loading ramps. Rails or pockets for the attachment of loading ramps shall be provided at the rear of the semitrailer platform for type I, class A (level deck, fixed gooseneck) semitrailer and, when specified (see 6.2), shall be provided at the rear for other types and classes.

* 3.7.9 Platform width extension. When specified (see 6.2), swing-type side extension brackets for planking shall be furnished to provide at least 1 foot of additional width to each side of the semitrailer. The number of brackets per side and design shall be the manufacturer's commercial standard, except in no case shall the bracket center to center spacing be more than 2 feet. Provisions shall be made for folding and locking the brackets against the main frame rails when the brackets are not being used. The brackets shall be so located, vertically, that when the removable planks specified in 3.7.1 are in place, the top of the planks shall be level with the remaining planks in the platform.

* 3.8 Gooseneck. The gooseneck shall be of sufficient length to provide the minimum landing wheel clearance (turning clearance) specified and narrow enough in width and shaped to meet the maximum forward swing clearance radius specified.

* 3.8.1 Class A, fixed gooseneck. A fixed, flat type gooseneck shall be provided on class A semitrailers. The entire area between the gooseneck main frame members shall be inclosed for use as a tool box. The underside shall be inclosed with a steel plate not less than 3/16 inch thick. Drain holes shall be provided. The top shall be covered with not less than three hinged deck lids, arranged to provide access to all storage space. The deck lids shall be of safety tread steel

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plate with a minimum thickness of 3/16 inch, exclusive of the raised tread. Ring or retractable handles shall be provided to swing open each deck lid. Suitable hasps shall be provided to permit padlocking the deck lids. Air lines and electrical cables and components within the storage area shall be protected from damage by stowed equipment but shall be accessible for servicing.

3.8.1.1 Class A fixed gooseneck with a full width gooseneck. When specified (see 6.2), the fixed gooseneck for class A shall be the full width of the loadspace. The top of the gooseneck outside of the tool box area shall be completely covered with a steel plate of the safety tread type with a minimum thickness of 3/16 inch exclusive of the raised tread; or, when specified (see 6.2), floored longitudinally, outside of, between and flush with the top of the frame with hardwood lumber (see 3.7.1) not less than 1-5/8 inches thick.

3.8.2 Class B, folding gooseneck. The gooseneck for class B shall be the full width of the loadspace and shall be of the folding type. The gooseneck and operating mechanisms shall be functionally compatible with a truck tractor's accommodating accessories, including its winch assembly, provided for raising and lowering the semitrailer gooseneck from the ground to fixed gooseneck travel position to mate with a tractor fifth wheel. The front edge of the gooseneck shall provide a rounded surface for compatibility with a truck tractor and for ease of loading tracked vehicles. A hinged connection between the load carrying portion and the gooseneck deck shall permit the gooseneck to be laid on the ground in a prone position, to form a continuous full width ramp and loading approach to the semitrailer deck. A locking arrangement shall be provided to enable the gooseneck to be held erect in a fixed gooseneck type position.

3.8.2.1 Lift fitting, class B, folding gooseneck. A heavy duty lift fitting, with removable pin for quick disconnect winch cable attachment shall be provided. The lift fitting shall be located at the center front of the gooseneck. The removable lifting pin shall be provided with a corrosion-resistant security chain or cable to prevent loss.

* 3.8.3 Class C, detachable gooseneck. The gooseneck for class C shall be of the detachable type. Throughout the detaching and attaching operation, the gooseneck always shall be locked to either a truck tractor or the semitrailer. The design shall enable the lowering and raising of the front end of the platform.

3.8.4 Class D, hydraulic folding gooseneck. The gooseneck for class D shall be the full width of the loadspace and shall be of the hydraulic folding type. The front edge of the gooseneck shall provide a rounded surface for compatibility with a truck tractor and for ease of loading tracked vehicles. A hinged connection between the load carrying portion and the gooseneck deck shall permit the gooseneck to be laid on the ground in a prone position, to form a

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continuous full width ramp and loading approach to the semitrailer deck. A locking arrangement shall be provided to enable the gooseneck to be held erect in a fixed gooseneck position.

3.8.4.1 Hydraulic folding gooseneck. Class D folding gooseneck semitrailer shall be furnished with a hydraulically operated folding gooseneck mechanism and a self-contained hydraulic power system. Folding and unfolding operations shall be capable of being performed without outside power or mechanical assistance. Means shall be provided to prevent damage to the kingpin with the gooseneck in the prone position. Any locking pins shall be of the captive type or shall be secured to the semitrailer with corrosion-resistant chains or cables. A self-contained hydraulic system shall be installed on the semitrailer.

3.8.4.2 Hydraulic system. The hydraulic system shall consist of an engine, hydraulic pump, hydraulic cylinders, hydraulic fluid reservoir, control and relief valves, filters, hoses, piping, fittings and all other components required to actuate and control the hydraulically powered gooseneck (and for class D, the landing legs, when landing legs are furnished). The cylinders shall be designed to travel slower in the lift mode than in the retract mode.

3.8.4.3 Hydraulic pump. The hydraulic pump shall be capable of delivering a flow rate at maximum system operating pressure which is adequate to provide for a minimum system flow rate of 3 gallons per minute (gpm).

3.8.4.4 Hydraulic fluid and filter. A full-flow hydraulic filter, having a filter rating no greater than 25 microns shall be installed in the hydraulic system. The hydraulic fluid shall be of a noncorrosive type with characteristics to withstand a temperature range from minus 40 to plus 275 degrees Fahrenheit (°F). The hydraulic fluid shall conform to SAE standards.

* 3.8.4.5 Hydraulic reservoir. The hydraulic reservoir shall have sufficient capacity to hold the maximum volume of fluid required for the system plus a minimum of 20 percent reserve. Further, in no case shall the reservoir capacity be less than four gallons. The reservoir shall be vented by a separate vent or through the filler cap. The vent shall be fitted with a filter element. A corrosion-resistant plate conforming to composition A (class l or 2) or composition C of MIL-P-514, indicating the type of hydraulic fluid to be used, shall be installed near the filler cap.

3.8.4.6 Hydraulic lines and fittings. All hydraulic tubing, hosing, and fittings shall conform to applicable SAE standards. A minimum number of fittings, joints, and connections shall be used to prevent excessive back pressure, clogging, and leakage. All hoses and fittings shall be capable of withstanding a bursting pressure of four times the working pressure and a maximum proof pressure of at least two times the working pressure. Hoses shall have physical qualities equal to or conforming to SAE J517 (100 R1), as applicable. Hoses shall be protected against chafing through utilization of supports and protective devices.

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* 3.8.4.7 Engine. Unless otherwise specified (see 6.2), an air cooled, four-stroke cycle gasoline engine shall be furnished to drive the hydraulic pump. When specified (see 6.2), an air cooled, diesel engine shall be furnished. The engine shall have adequate horsepower, torque and speed characteristics to meet the performance requirements of 3.8.4.2. The engine shall be furnished complete with all accessories required for operation including a fuel tank with not less than a 3-gallon capacity; a 12-volt starting battery of the maintenance-free type, having the maintenance-free characteristics listed in W-B-131; a battery charging alternator; a fuel indicator, air cleaner, fuel gage, battery charging indicator, throttle, choke (gasoline engine), engine starting and stopping controls and all other components required to power the hydraulic system. The oil drain shall be located so that the system can be drained, without spillage, into a container sitting on the ground.

3.8.4.8 Radio interference suppression. The engine shall be suppressed to limit electromagnetic radiation in accordance with SAE J551.

* 3.8.4.9 Engine sound level. The engine noise level during operation shall be not more than 84dB(A) when measured in accordance with MIL-STD-1474. The noise level shall be measured at the operator's station adjacent to the controls with the operator standing on the ground.

* 3.8.4.10 Landing gear for semitrailers with a hydraulic folding gooseneck. The semitrailer shall have two hydraulically operated landing legs or fixed landing pads. When landing legs are furnished, they shall be equipped with self-leveling landing pads not less than 100 square inches in area. The landing legs shall be capable of being hydraulically extended to the operating positions and hydraulically retracted to allow the gooseneck to fold to the ground as specified in 3.8.4 with full payload, without any evidence of any type of component failure, deformation, permanent set or misalinement. The landing gear shall withstand, without deformation, the combined static and dynamic forces due to the proportion of the gross weight sustained and the forces resulting from impact during coupling and uncoupling operations. When placed in travel position, the landing gear legs shall remain locked. The landing gear shall be protected to prevent the entrance of foreign matter which would impair its functional or mechanical efficiency.

3.8.4.11 Range of adjustment for class D folding gooseneck landing gear. The landing gear shall have a range of adjustment sufficient to vary the height of the upper fifth wheel from 47 inches to not less than 53 inches from the ground for sizes 15, 20, 25 and 35 and from 52 inches to not less than 58 inches from the ground for sizes 50, 60, and 75. Sizes 50, 60 and 75 shall be able to raise the folding gooseneck off of the fifth wheel on truck tractors with a fifth wheel height as great as 58 inches. With the semitrailer coupled to a towing tractor and in a level position, the clearance under the fully retracted landing gear for all sizes, shall exceed the semitrailer ground clearance and further, shall be not less than 11 inches.

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3.8.4.12 Landing pads for class D folding gooseneck landing gear. When landing pads are furnished in lieu of landing legs, they shall be of the fixed type, of adequate strength and surface area to support the loaded semitrailer and adequate to sustain all loads and forces during loading and unloading operations. The landing pads shall be provided at the rear of the gooseneck and at the front of the main platform and shall have a ground clearance of not less than 12 inches with the gooseneck in travel position and the main platform level. The landing pads, in conjunction with the hydraulic system, shall enable the fifth wheel height to be varied as specified in 3.8.4.11.

3.8.4.13 Gooseneck and landing gear controls for class D. Controls shall be within easy reach of the operator standing on the ground. The controls shall be located on the left side (street side) of the gooseneck.

* 3.8.5 Class E, hydraulic detachable gooseneck. The gooseneck for class E shall be of the hydraulically operated detachable type. Throughout the detaching and attaching operation, the gooseneck always shall be locked to either a truck tractor or the semitrailer. The design shall enable the lowering and raising of the front end of the platform. A self-contained engine driven hydraulic system to operate the hydraulic cylinder(s) during connect and disconnect operations, independent of a truck tractor, shall be furnished on the detachable gooseneck. The system shall be mounted where it will receive maximum protection from road shocks and heavy vibrations. The system shall conform to 3.8.4.2 through 3.8.4.9.

3.9 Axles and suspension. The axle and suspension system shall be of rated capacity at least equal to the load imposed at the ground when the semitrailer is loaded with its specified payload. Clearances shall prevent interference between wheels and any other portion of the vehicle under the conditions specified for each style. Differences in height for specified sets of dual wheels shall be measured at the center of the tread on the outer tire. The distance between the centerlines of adjacent axles shall be not less than 50 inches.

3.9.1 Style 4 system. Style 4 shall consist of a spring suspension system for type V level deck extended semitrailer and of either the walking beam or spring suspension, or combination spring and walking beam suspension, for all other types, and shall mount four dual wheels. Sufficient articulation shall be provided to maintain loading on each wheel, with the semitrailer loaded, under each of the following conditions:

(a) With not more than two diagonally opposite wheels elevated 6 inches
(b) With the forward pair of dual wheels elevated 6 inches above the rear pair
(c) With the rear pair of dual wheels elevated 6 inches above the forward pair.

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3.9.2 Style 6 system. Style 6 shall consist of a combination spring and walking beam suspension and shall mount six dual wheels. Sufficient articulation shall be provided to maintain loading on each wheel, with the semitrailer loaded, under each of the following conditions:

(a) With the first and third wheels on one side and the center wheel on the opposite side elevated 6 inches above the remaining wheels
(b) With any pair of opposite wheels elevated 6 inches above the remaining wheels
(c) With any two pairs of opposite wheels elevated 6 inches above the remaining pair.

3.9.3 Style 8 system. Style 8 shall consist of two independent walking beams, pivoting about a common transverse centerline, each mounting two trunnion axles with dual wheels on each side of the beam. Sufficient articulation shall be provided to maintain loading on each wheel, with the semitrailer loaded, under each of the following conditions:

(a) With the outside pairs of dual wheels resting on surfaces 6 inches higher or 6 inches lower than the inner pairs of dual wheels
(b) With the forward wheels on each beam elevated 6 inches above the rear wheels
(c) With the rear wheels on each beam elevated 6 inches above the forward wheels.

3.9.4 Oil lubricated axles. Except for Style 8W (which may be oil lubricated at the manufacturer's option), the wheel bearings and axle spindles shall be oil lubricated. The oil viscosity shall be in accordance with the manufacturer's recommendations. The hubs shall be fitted with seals. The hub caps shall have a window and shall incorporate provisions for replenishing the oil supply. The hub caps shall incorporate provisions for venting or an equivalent method of withstanding pressures without leakage.

* 3.9.5 Air suspension system. When specified (see 6.2), the semitrailer shall be equipped with the manufacturer's standard or optional air suspension system. Components shall have a rated capacity at least equal to the load imposed on each member, measured at the ground, with the semitrailer loaded as specified in 3.4. Hydraulic double-acting shock absorbers shall be provided on each axle assembly. The suspension system shall incorporate leveling valves with time delay to avoid constant air consumption. Shock absorbers and springs shall be fabricated of materials capable of operating in ambient temperatures from minus 50 degrees Fahrenheit (°F) to plus 125°F. No other major spring medium, such as leaf, coil, or rubber springs, shall be used in conjunction with the air springs. Suspension arms shall be equipped with rubber bushings that do not require periodic lubrication. The suspension system shall be provided with a mechanism at each wheel to assure lifting of the wheel and axle when jacking the semitrailer from the applicable jacking location.

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* 3.10 Wheels, rims, tires and tubes. Rims, and tire ratings, shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations, for the type and size of tires furnished and for operation under the conditions specified in table II. Except when tube type tires are specified in acquisition documents, multi-piece rims shall not be furnished on single rear axles rated at 23,000 pounds or less and on rear tandem axles rated at 52,000 pounds or less. Tire, rim size, disc wheel size, and ply rating shall be the same for all wheels on each vehicle. When specified (see 6.2), disc type wheels shall be furnished.

* 3.10.1 Tires. Tires shall be of the steel belted radial type. Tires shall have highway tread. Tires shall be of the tubeless type or, on axles where multi-piece rims are permitted (see 3.10), may be of the tube type. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the semitrailer loaded with specified payload. Tires and tire size designation systems shall conform to the Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations. The number of tires furnished on each semitrailer shall be eight for style 4, twelve for style 6 and sixteen for style 8, plus any spare tires required.

3.10.2 Inner tubes. When tube-type tires are permitted herein or specified in acquisition documents, inner tubes shall be of the heavy duty type, and shall be of the proper size for the tires furnished. Tire flaps shall be provided for tube-type tires in accordance with Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations.

3.10.3 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be furnished. The spare wheel or rim shall be the same size as those furnished on the semitrailer. The spare shall be securely stowed on the vehicle for shipment only.

3.10.4 Spare tire. When specified (see 6.2), an inflated spare tire, mounted on the spare wheel or rim, shall be furnished. The spare tire shall be of the same size, tread design and load range (ply rating) as the tires furnished on the vehicle. The spare assembly shall be securely stowed on the vehicle for shipment only.

* 3.10.5 Spare tire storage compartment. When specified (see 6.2), an anti-theft spare tire storage compartment shall be furnished. The storage compartment shall be constructed of not less than 12 gage sheet steel and shall be bolted to frame structural members. A hinged lid or door and provisions for a padlock shall be furnished. The compartment shall neither impair the use of the loading surfaces nor reduce the vehicle ground clearance.

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* 3.10.6 Spare tire carrier. When specified (see 6.2), for type V level deck extended, a spare tire carrier shall be installed under the platform at the curbside in an accessible location. Means shall be provided for securing the tire within the carrier to prevent accidental loss. Threaded fasteners, when used to secure the spare assembly in the carrier, shall be constructed of or plated with corrosion-resistant material.

* 3.11 Brakes. The brakes shall be of the full-air internal expanding type and shall conform to Federal Motor Carrier Safety Regulations 393.40, 393.43, and 393.45 through 393.52. Brake linings shall be of nonasbestos material. The brake system shall include automatic slack adjusters, piping, hose connections, gladhands, spring loaded dust covers or dummy gladhands equipped with security chains or cables, air reservoirs (with remote operating control for the drain valve positioned above the bottom of the frame rail), brake chambers, relay emergency valve, and all other parts and connections required for a complete air-brake system. Gladhands shall conform to SAE J318. Location of air fittings shall be as recommended in SAE J702, except on class B and class D folding gooseneck semitrailers, fittings shall be located near the front of the gooseneck and shall be recessed in the gooseneck to prevent damage. All components of the braking system shall be installed in such a manner as to provide adequate road clearance when traveling over uneven or rough terrain, including objects liable to strike and cause damage to the brake system components. No part of the braking system shall extend below the bottom of wheel rims, to insure, in case of a flat tire, that the weight of the semitrailer will be supported by the rim and the flat tire and not be imposed on any component of the braking system. Slack adjusters and air chambers shall be located above the bottom edge of the axle carrier.

* 3.11.1 Parking brakes. The semitrailer shall be equipped with spring or air diaphragm mechanical lock type parking brakes. The parking brakes shall hold the semitrailer, with rated payload, on a 10 percent grade despite the depletion of the compressed air supply. With no additional application, the parking brakes shall hold the semitrailer, with rated payload, on an 10 percent incline with the semitrailer headed first up and then down the incline. The parking brakes shall be automatically applied upon disconnection of the emergency air line and under emergency conditions. The parking brakes shall conform to Federal Motor Carrier Safety Regulation 393.41.

* 3.12 Upper fifth wheel. The upper fifth wheel plate shall completely cover a 36 inch diameter lower fifth wheel when the lower fifth wheel is coupled to the kingpin of the upper fifth wheel and the semitrailer is turned at all angles. The upper fifth wheel shall conform to Federal Motor Carrier Safety Regulation 399.70(b).

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* 3.12.1 Upper fifth wheel plate. The upper fifth wheel plate shall be not less than 1/4 inch thick for semitrailers up to and including size 20 and not less than 3/8 inch thick for all larger sizes. The plate shall be reinforced on the upper surface through the use of commercial steel angles, channels, and ribs. The plate shall include an access hole to permit servicing the brake connections and electrical connectors when these components are not serviceable from the top.

* 3.12.2 Pickup plate. A turned up lip pickup plate, compatible with the size (payload) of the semitrailer, shall be provided as an extension of the upper fifth wheel plate. The pickup plate shall have a turned up lip not less then 2 inches high. The pickup plate shall extend sufficiently forward to protect the semitrailer from damage during coupling and uncoupling operations to and from a truck tractor. The pickup plate shall be integral with the upper fifth wheel plate.

* 3.12.3 Kingpin. A 2-inch diameter kingpin conforming to SAE J700 shall be furnished. The kingpin shall be fabricated of heat treated alloy steel.

* 3.13 Landing gear, class A. Unless otherwise specified (see 3.13.3), elevating landing gear shall be furnished on class A fixed gooseneck semitrailer. The landing gear shall consist of two legs of the vertically retractable, two-speed type and shall be controlled by means of a hand crank(s). Each leg shall be equipped with steel landing pads not less than 10 inches square. Landing legs shall be located on each side at the rear of the gooseneck. Each leg shall be capable of supporting its portion of the fully loaded semitrailer and shall be adequately braced to the lowest extremity of the fixed portion of each leg. Supports for the crank extension shafts and clips for holding the cranks when folded shall be provided. The landing gear shall be protected to prevent the entrance of foreign matter which would impair its functional or mechanical efficiency. On size 15, 20, 25 and 35 semitrailers, the maximum cranking effort to raise the fully loaded semitrailer shall be not greater than 100 pounds. On size 50, 60 and 75 semitrailers, the maximum cranking effort to raise the empty semitrailer shall be not greater than 100 pounds.

* 3.13.1 Landing gear range of adjustment, class A. The landing gear shall have a range of adjustment sufficient to vary the height of the upper fifth wheel plate from 47 inches to not less than 53 inches from the ground for size 15, 20, and 25 semitrailers and from 52 inches to not less than 58 inches from the ground for size 35, 50, 60 and 75 semitrailers. The minimum ground clearance with the landing gear retracted shall be not less than the semitrailer frame ground clearance.

* 3.13.2 Air powered landing gear. When specified (see 6.2), in addition to the requirements of 3.13 and 3.13.1 (less hand crank, etc.), the landing gear shall be air powered. The air powered landing gear system shall include a control box equipped with a 12-volt direct current (dc) motor driven pump, master control valve, leveling valve, booster, main oil tank, air side oil tank, key switch and mounting hardware.

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* 3.13.3 Landing supports, class A. When specified (see 6.2), landing supports shall be furnished in lieu of landing gear. Landing supports shall consist of two legs of the gravity telescopic type. Each leg shall be equipped with a steel landing pad not less than 10 inches square. Minimum ground clearance with landing supports retracted shall be not less than the semitrailer frame ground clearance. Landing supports shall have a range of adjustment sufficient to vary the height of the upper fifth wheel plate from 47 inches to not less than 53 inches from the ground for size 15, 20, and 25 semitrailers and from 52 inches to not less than 58 inches from the ground for size 35, 50, 60 and 75 semitrailers. Landing support adjustment shall be in not more than 2 inch increments.

* 3.14 Skid blocks, class B. On class B folding gooseneck semitrailers, skid blocks shall be provided in lieu of landing gear. Skid blocks shall be permanently attached under the front end of the platform. Skid blocks shall be of adequate size to prevent the loaded semitrailer from sinking into soft ground. With the semitrailer platform in a level position with load, the minimum ground clearance under the skid blocks shall be not less than the semitrailer frame ground clearance, minus 1/2 inch.

3.15 Loading ramps. When specified (see 6.2), two loading ramps of all metal construction, complete with hooks and handles, shall be provided. The ramps, when properly blocked at the midpoint, shall be capable of withstanding an equally applied portion of the full semitrailer payload without permanent deformation. Ramps shall be of a length to provide a ramping angle of not less than 12 degrees and not more than 15 degrees.

* 3.15.1 Fold up loading ramps. When specified (see 6.2), two permanently attached, fold up, all metal loading ramps shall be furnished. The ramps shall be hinged to the rear of the semitrailer. The ramps shall be capable of sliding from the extreme outside edges of the semitrailer to within 12 inches of the centerline of the semitrailer while still hinged and attached. The ramps shall be provided with devices to secure them in travel position. The ramps, when properly blocked at the midpoint, shall be capable of withstanding an equally applied portion of the full semitrailer payload without permanent deformation. Ramps shall be of a length to provide a ramping angle of not less than 12 degrees and not more than 15 degrees.

* 3.16 Lubrication system. All moving parts normally requiring lubrication shall be provided with accessible pressure grease fittings. Pressure type lubrication fittings shall be readily accessible for servicing and shall conform to SAE J534. Pressure relief fittings shall be used where high pressure lubricating equipment could cause damage to seals.

* 3.17 Lighting. All lights and reflectors shall be protected from operational hazards by mounting in recessed or otherwise guarded locations. The brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Lights and reflectors shall not be mounted on vertical surfaces of the rub rail (unless recessed and fully

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protected) or on the semitrailer bumper. Turn signal lamps shall conform to SAE J588. All electrical wiring shall conform to SAE J1292. When specified and only when specified (see 6.2), clearance, side marker and identification lamps shall be constructed for easy removal and replacement of lamps and lenses without the use of any tools. Otherwise, if the option is not specified, the use of tools shall be required to minimize pilferage of lamps and lenses.

3.17.1 Twelve-volt dc system. A 12-volt dc lighting system conforming to 3.17 and 3.17.2 shall be provided. The 12-volt dc lighting system shall conform to Federal Motor Carrier Safety Regulations 393.14, 393.20, 393.22, 393.23, 393.25 through 393.29, 393.32 and 393.33.

3.17.2 Receptacle for 12-volt dc system. The front of the semitrailer shall be equipped with a 7-contact receptacle conforming to SAE J560 with the connectors connected and color coded as specified therein. The receptacle shall be provided with a spring loaded cover. The receptacle shall be located in accordance with SAE J702.

3.17.3 Independent 24-volt dc system. When specified (see 6.2), a 24-volt dc lighting system, independent of the 12-volt dc lighting system, shall be provided. Separate wiring and bulbs shall be provided for the 24-volt dc system. The system shall conform to 3.17. The 24-volt dc system shall also conform to the Federal Motor Carrier Safety Regulations referenced in 3.17.1, except the 24-volt dc stop and turn lamps shall be combined.

3.17.4 Interconnected 24-volt dc system. When specified (see 6.2), capability of operating the normal 12-volt dc system (see 3.17.1) from a towing vehicle equipped with a nominal 24-volt dc electrical system shall be provided.

3.17.5 Receptacle for 24-volt dc system. When either an independent or interconnected 24-volt dc system is specified, the front of the semitrailer shall also be equipped with a 12-contact receptacle and cover conforming to MS75021, part number MS75021-1. The receptacle shall be located in accordance with SAE J702. The 24-volt dc, 12-contact receptacle shall be connected to the lights as follows:

Contact B.	Connect to left-hand turn signal and stop lamp (yellow)
Contact D.	Connect to ground (white)
Contact E.	Connect to clearance, side marker, identification, and tail lamps (black and brown)
Contact J.	Connect to right-hand turn signal and stop lamp (green).
Contact L.	Connect to ground (white).

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The remaining contacts shall not be connected. Circuits B and J on tactical (military design) trucks are combination stop and turn indicator circuits. On the interconnected 24-volt dc system, the normal 12-volt dc turn signal lights will function both as turn signals and stop lights, and the normal 12-volt dc stop lights will not be operational when the semitrailer is connected to a towing vehicle with a 24-volt dc power supply. Because of this condition, the stop light (red) circuit is not connected to the 24-volt, 12-contact receptacle.

3.17.6 Resistors. When an interconnected 24-volt dc system is specified (see 3.17.4), the 12-contact receptacle shall be provided with resistance in each circuit to reduce the voltage of a tactical towing vehicle from a nominal 28 volts dc to 12 volts dc. Each circuit resistor shall be selected to reduce the voltage of the tactical truck to within the maximum rated voltage of the semitrailer electrical components. The resistor assembly shall be located in a protective housing and provided with adequate ventilation or a heat sink to prevent overheating and any heat damage to resistors, wiring, or adjacent components.

* 3.18 Mud flaps. Mud flaps shall be provided to the rear of the rear wheels, in accordance with SAE J682. A metal strip not less than 1/8 inch thick and not less than one inch wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 1 inch high by 1-1/4 inch wide by 3/32 inch thick shall be furnished at each bolt.

* 3.19 Rear end protection. The semitrailer rear end protection shall be in accordance with Federal Motor Carrier Safety Regulation 393.86. When specified (see 6.2), a heavy duty reinforced bumper plate shall be provided at the rear for protection against damage when backing and for use when the semitrailer is pushed from the rear to assist in traveling over unimproved roads. The bumper shall withstand the forces imposed during pushing of the fully loaded semitrailer under severe operating conditions without damage to structural members, lights, reflectors or other components.

* 3.19.1 Push bumper. On class B folding gooseneck semitrailers, a heavy duty reinforced bumper plate, centrally located and approximately flush with the rear of the semitrailer, shall be provided. The bumper plate shall allow truck tractor bumper contact during disabled vehicle winch loading operations, without damage to the truck tractor or semitrailer.

3.20 Servicing and adjusting. Prior to acceptance by the Government inspector, the contractor shall service and adjust the semitrailer for operational use including at least the following: adjustment of electrical and brake systems; inflation of all tires; and complete lubrication with grades of lubricants recommended for the ambient air temperature at the delivery point.

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3.21 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the best commercial practice.

3.21.1 Welding. Welding shall conform to the American Welding Society's Structural Welding code, Steel, D1.1 and, when specified (see 6.2), to the requirements of MIL-STD-1261, class 2. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welding rods shall be compatible with the metal being welded. Spot, tack or intermittent welds shall not be used where structural strength is required. Weld penetration shall provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration.

3.21.1.1 Welding workmanship specimens. When specified (see 6.2), the contractor shall prepare, or shall have prepared, welding workmanship specimens in accordance with the requirements of MIL-STD-1261, class 2 representing each typical weld joint encountered on the semitrailer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performing of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor shall be subject to Government verification at unscheduled intervals. Verification shall consist of observation of the operations to determine that practices, methods, and procedures of the

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contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. When specified (see 3.21.1.1), weld specimens shall be used by as a guide in determining the quality of all applicable welds. All welds on the first production vehicle and on all subsequent production vehicles under the contract shall be compared with applicable weld workmanship specimens prior to priming or painting the vehicle. Acceptance of the first vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.3.1 Semitrailer weight. The semitrailer shall be weighed to determine curb weight and distribution of curb weight on the kingpin and the rear axles. The imposed loading on the kingpin and rear axles, with payload, shall be computed using curb weight and the semitrailer payload distributed as specified in 3.3. The rated capacity of the axles, suspension system and tires shall be compared to the loads imposed to determine if they conform to contract requirements.

* 4.3.2 Normal road test. Unless an extended road test is specified (see 4.3.3), the first vehicle for each contract line item number (CLIN) under contract; except sizes 50, 60 and 75 (see 4.3.10); shall be road tested with and without payload. The road test shall be for the distance required in table V, corresponding to the number of vehicles being procured under the specific contract line item number (CLIN). The test shall be conducted by the contractor under the direction of Government representatives. The test shall be conducted over a test course approved by the procuring activity and meeting the requirements of table VI. The vehicle shall be inspected periodically during and at the completion of the test by the contractor.

TABLE V. Road test mileage.

Number of vehicles being procured	Mileage of road test
1 - 10	100
11 - 20	200
21 - 30	300
31 - 40	400
41 - 50	500
51 - 100	1000
100 - up	5000

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TABLE VI. Test course requirements.

Type of road	Percent of miles
Dirt	10
Gravel	30
Improved pavement	60

* 4.3.2.1 Test course. The total test mileage shall be on roads and highways within the general geographic limits of the place of manufacture and adjoining states. Any special highway permits required shall be obtained by the contractor. The course shall be arranged to return the vehicle to the manufacturer's plant at the completion of each 1,000 miles, if applicable, or at the completion of the test otherwise, for inspection and chassis servicing.

4.3.2.2 Test loads. Type I level deck and type II drop deck semitrailers shall be tested for 90 percent of the mileage with the semitrailer loaded as specified in table III and for 10 percent of the mileage with the semitrailer empty. Type V level deck extended semitrailers shall be tested for 45 percent of the mileage with the rated load evenly distributed, for 45 percent of the mileage with 65 percent of rated load concentrated as specified in 3.3, and for 10 percent of the mileage empty. Average speeds during all road tests shall be as close as possible to those specified in table II.

* 4.3.3 Extended road test. When specified (see 6.2), the first vehicle shall be subjected to an extended road test as specified in table VII. While negotiating the specified side slopes, in both directions, the equipment shall be subjected to not less than 10 sine wave turns. While negotiating the paved road course at highway speed (see table II), the equipment shall be periodically subjected to at least 25 sudden stops at maximum braking effort and shall be periodically subjected to at least 25 left and 25 right turn rapid lane change maneuvers that simulate avoidance of collisions.

TABLE VII. Extended road test.

Surface	Load condition	Average speed (mph)	Distance (miles)
Side slope, 15 percent	Full load	5 – 10	
	Half load	5 – 10	
	Empty	5 – 10	
Belgian block	Full load	10 – 20	300
	Half load	10 – 20	200
	Empty	10 – 20	100

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TABLE VII. Extended road test. - (continued)

Surface	Load condition	Average speed (mph)	Distance (miles)
Paved road	Full load	Highway	2100
	Half load	speed (see	1400
	Empty	table I)	700
Improved gravel road	Full load	25 - 35	900
	Half load	25 - 35	600
	Empty	25 - 35	300

* 4.3.3.1 Patterning. High stress areas of the frame assembly shall be coated with brittle lacquer prior to the road test. The semitrailer shall be fully loaded and towed over the Belgian block test course for one cycle.

* 4.3.3.2 Strain recording. Strain gage rosettes shall be positioned in locations of high stress concentrations. The locations shall be as determined by the brittle lacquer patterning. A minimum of ten strain gages shall be used. Locations of strain gages shall include the gooseneck, the kingpin area and transverse areas. Strains shall be recorded for each of the following three cycles of the test course:

(a) Full load 1 cycle
(b) 40 to 60 percent of full load 1 cycle
(c) No load 1 cycle

Strain/occurence histograms shall be supplied for each strain gage for each test cycle. The actual test course shall be recorded on the histogram. Pictures indicating the position of each strain gage shall be provided.

* 4.3.3.3 Evaluation. The ultimate, tensile and yield strengths of the gaged components shall be documented. Any strain recorded indicating stress greater than that specified in 3.6 shall be cause for rejection.

* 4.3.3.4 General design and performance characteristics verification. The equipment shall be verified as suitable for the intended use by satisfactorily completing the preceding inspections and tests. Equipment submitted for test shall contain all specified components and shall be properly adjusted and serviced for operation. Apparatus used in conjunction with the tests shall be calibrated for accuracy. In addition to the results of inspections and tests, test records shall include the following information, as applicable.

(a) Name, date(s), and location of test
(b) Description of test course and weather conditions that could affect test results

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(c) Condition of the equipment at the start of test
(d) Tire inflation pressure
(e) Types of lubricants used
(f) Description of test instrumentation and recording apparatus, including calibration records
(g) Description of the truck tractor used
(h) Load condition of the equipment
(i) Average and maximum speeds
(j) Miles traveled, including total test mileage and accumulated mileage, and the time of each major incident such as component failure or adjustment
(k) Component failure analysis and corrective measures
(l) Component adjustment, and method and ease of adjustment
(m) Any noted safety hazards
(n) Condition of the equipment at the conclusion of the test.

* 4.3.3.5 Reliability and maintainability report. A report pertaining to reliability and maintainability shall be furnished after completion of the extended road test. The report shall contain the following information:

(a) All failures, maintenance, and other events shall be recorded and identified by accumulated operating time, miles, cycles, or position in the test procedure as appropriate. Test conditions during the failures or irregular operations identified shall be recorded.
(b) Description of the engineering reasoning, and of any tests conducted to determine assignable causes for all failures or irregular operations identified.
(c) Description of the engineering reasoning behind any proposed corrections to be made on the production items, and the predicted effectiveness of those corrections.
(d) Test activity or contractor comments on item, features or requirements that, if modified, should improve the item.
(e) Estimates (within 5 percent of actual experienced) of manhours required for each maintenance and servicing action during the test. A brief description of the qualifications and experience of the personnel involved shall be included and shall be adequate to allow comparison to the personnel anticipated in similar field work.

* 4.3.4 Turning and tracking tests. During the road test, conformance to turning and tracking requirements (see 3.3.1 and 3.3.3) shall be verified.

* 4.3.5 Landing gear test. Periodically during the road test, functions of the landing gear shall be tested to verify conformance to 3.13.

4.3.6 Brake tests. Brake tests to determine conformance to 3.3.2 shall be conducted during the road test. The brake tests shall be conducted with the fully loaded semitrailer coupled to a truck tractor. The truck tractor shall have a gross vehicle weight rating (GVWR) and gross combination weight rating (GCWR) not exceeding that specified in table VIII, and shall be

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without excessive brake capacity. GVWR and GCWR shall be the maximum ratings on the tractor only and not necessarily the actual maximum GVW and GCW of the combination of test vehicles. Failure of the vehicle combination to stop within the required 40 feet from a speed of 20 mph shall be cause for rejection.

TABLE VIII. Maximum tractor characteristics.

Size of semitrailer	Max. tractor GVWR, pounds	Max. tractor GCWR, pounds
15	26,000	50,000
20	28,000	60,000
25	35,000	75,000
35	50,000	100,000
50	70,000	137,000
60	77,000	170,000
75	100,000	200,000

* 4.3.7 Suspension system articulation test. The fully loaded semitrailer shall be driven onto 6 inch blocks to simulate each of the conditions in 3.9.1, 3.9.2 or 3.9.3, as applicable. Failure of any wheel to maintain contact with the ground or the blocks shall be cause for rejection.

4.3.8 Electrical test. An electrical test shall be conducted to determine that all vehicle lights are functioning. In addition, when an interconnected 24-volt system is furnished, the circuits of the 12-contact receptacle shall be energized with not less than 28 volts dc regulated voltage for a time sufficient to stabilize the temperature of the resistor assembly for not less than 30 minutes in an ambient temperature of not less than 77°F. After the test, the resistor assembly, including resistors, wiring and adjacent components shall be examined for evidence of overheating, deterioration or damage.

4.3.9 Folding and detachable gooseneck test. The gooseneck on class B, C, D and E semitrailers shall be tested to determine conformance to the requirements specified herein (see 3.3.4, 3.8.2, 3.8.3, 3.8.4 and 3.8.5).

4.3.10 Load test for sizes 50, 60 and 75. Unless an extended road test is specified (see 4.3.3), sizes 50, 60 and 75 first vehicle shall be loaded with one and one-half times the rated payload and shall be examined to determine conformance to the static requirements of this specification. After completion of static examination, sizes 50, 60 and 75 first vehicle shall be road tested on public roads or may be road tested at the contractor's facility for a distance of not less than 5 miles, without payload, to determine conformance to the functional requirements of this specification.

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4.3.11 Test failure. The vehicle shall successfully complete the entire test. Failure of the vehicle to successfully complete the test shall be cause for nonacceptance of any of the contract quantity, pending correction of deficiencies and evidence of corrective action which shall prevent the recurrence of similar deficiencies in the contract quantity. Failure of the vehicle to successfully complete the test shall not constitute an excusable delay in meeting scheduled deliveries. The equipment shall be closely monitored throughout the test for interference with towing operations, and for instability as evidenced by poor tracking or uncontrollable sideslipping, skidding, swerving, or tilting. The equipment also shall be closely monitored for loosening of parts, interference between parts, leakage of fluids or lubricants, overheating of components, and for damage, permanent distortion, or excessive wear of parts and components. The equipment shall be disassembled to the extent necessary for inspection for unusual wear or damage to components. Substitution of a new semitrailer or replacement of a major component may require complete retest at the discretion of the Government representatives. Rejection of the test semitrailer shall be for damage or deficiencies, including but not limited to the following:

(a) Damage caused by collision
(b) Failure of any major component
(c) Vibration due to misalinement of wheels or frame
(d) Vibration due to the type of construction or mounting
(e) Evidence of abnormal tire wear due to misalinement or unbalance
(f) Failure of any vehicular safety device such as brakes or electrical circuits
(g) Evidence of structural weakness in any part of the vehicles, vehicle components, accessories or welds
(h) Loose mountings of parts or accessories due to workmanship or vehicular operation.

* 4.3.12 Mileage and maintenance records. The vehicle shall be driven not more than 50 miles in addition to the mileage required by 4.3.2, 4.3.3 or 4.3.10, as applicable. All mileage accumulated and all service or maintenance records shall be furnished to the Government representatives. Distribution of records shall be made identical to that for service manuals, and shall be concurrent with the manual distribution.

4.3.13 Delivery of test vehicle. Upon successful completion of the vehicle test and final examination, the vehicle shall be cleaned, serviced, and refurbished to the extent required for delivery of a new vehicle to the Government. Warranty provisions of the contract shall be exclusive of total test miles.

4.3.14 Production sample. Upon acceptance of the first vehicle, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of

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the contract. Release of the first vehicle for shipment prior to completion of the contract shall be at the discretion of the local Government Quality Assurance Representative (QAR).

4.3.15 Failure. Failure of the test vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under the contract until corrective action has been taken.

4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting the performance requirements contained herein.

4.5 Wood treatment certification. Manufacturer's records, acceptable to the Government, shall be furnished to verify that all wood requiring treatment in accordance with MIL-STD-1223 has been treated.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general nontactical use in transporting tracked and wheeled construction and material handling equipment, other tracked and wheeled vehicles, machinery, and general cargo.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, size, and style of semitrailer required (see 1.2).
(c) Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Identification of the appropriate service for painting and marking (see 3.1.1.1 and 3.1.1.2).
(e) Rustproofing, if required (see 3.1.1.3).
(f) Lifting and tiedown attachments, if required (see 3.1.1.6).
(g) Lubrication chart, if required (see 3.1.1.7).
(h) Whether 25 or 40 foot level load space is required for type V level deck extended semitrailer (see 3.5(a)).
(i) Frame strength at 33 percent of yield strength, if required (see 3.6).
(j) Platform extension (beavertail), if required (see 3.7.5).

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(k) Stake pockets, if required for type I level deck and type II drop deck (see 3.7.6).
(l) If stake pockets are not required for type V level deck extended (see 3.7.6).
(m) Loading ramp rails or pockets, if required for other than type I, class A (level deck, fixed gooseneck) (see 3.7.8).
(n) Platform width extension brackets, if required (see 3.7.9).
(o) Full width gooseneck, if required (see 3.8.1.1).
(p) Gooseneck area wood floor, if required (see 3.8.1.1).
(q) Pump diesel engine in lieu of gasoline, if required (see 3.8.4.7).
(r) Air suspension system, if required (see 3.9.5).
(s) Disc type wheels, if required (see 3.10).
(t) Spare wheel or rim, if required (see 3.10.3).
(u) Spare tire, if required (see 3.10.4).
(v) Spare tire storage compartment, if required (see 3.10.5).
(w) Spare tire carrier, if required for type V level deck extended (see 3.10.6).
(x) Air powered landing gear for class A, fixed gooseneck, if required (see 3.13.2).
(y) Landing supports, if required in lieu of landing gear for class A fixed gooseneck (see 3.13.3).
(z) Loading ramps, if required (see 3.15).
(aa) Fold up loading ramps, if required (see 3.15.1).
(ab) If lamp lenses and bulbs are to be removable without tools, see 3.17).
(ac) Independent 24-volt dc system, if required (see 3.17.3).
(ad) Interconnected 24-volt dc system, if required (see 3.17.4).
(ae) Welding conformance to MIL-STD-1261, if required (see 3.21.1).
(af) Welding workmanship specimens, if required (see 3.21.1.1).
(ag) Extended road test, if required (see 4.3.3).

* 6.3 Subject term (key word) listing.

Air suspension
Fifth wheel
Gooseneck
Hydraulic system
Landing gear
Semitrailer
Welding.

6.4 Deletions. Type III, low drop deck; and type IV, tilt deck; have been deleted from this specification. Tilt deck type semitrailers are covered in MIL-S-62101.

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6.5 Changes from previous issue. The margins of this document are marked with an asterisk (*) and underlined to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the previous issue.

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